

HPC GATEWAY: INFRASTRUCTURE AND CONSULTING FOR ACADEMIA AND INDUSTRY

BUILDING BRIDGES 2025

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Date: September 11, 2025



Agenda

- HPC Gateway – Bridging Research and Industry:
 - general introduction, ecosystem, pilot project
- CFD for Process Optimization:
 - Challenges and our recommender system approach
- Tomography and AI-based Denoising:
 - Ultrafast CT (ROFEX) at HZDR
 - Using machine learning to improve noisy data
 - Impact on science & industry applications

HPC GATEWAY – BRIDGING RESEARCH AND INDUSTRY

GENERAL INTRODUCTION, ECOSYSTEM, PILOT PROJECT

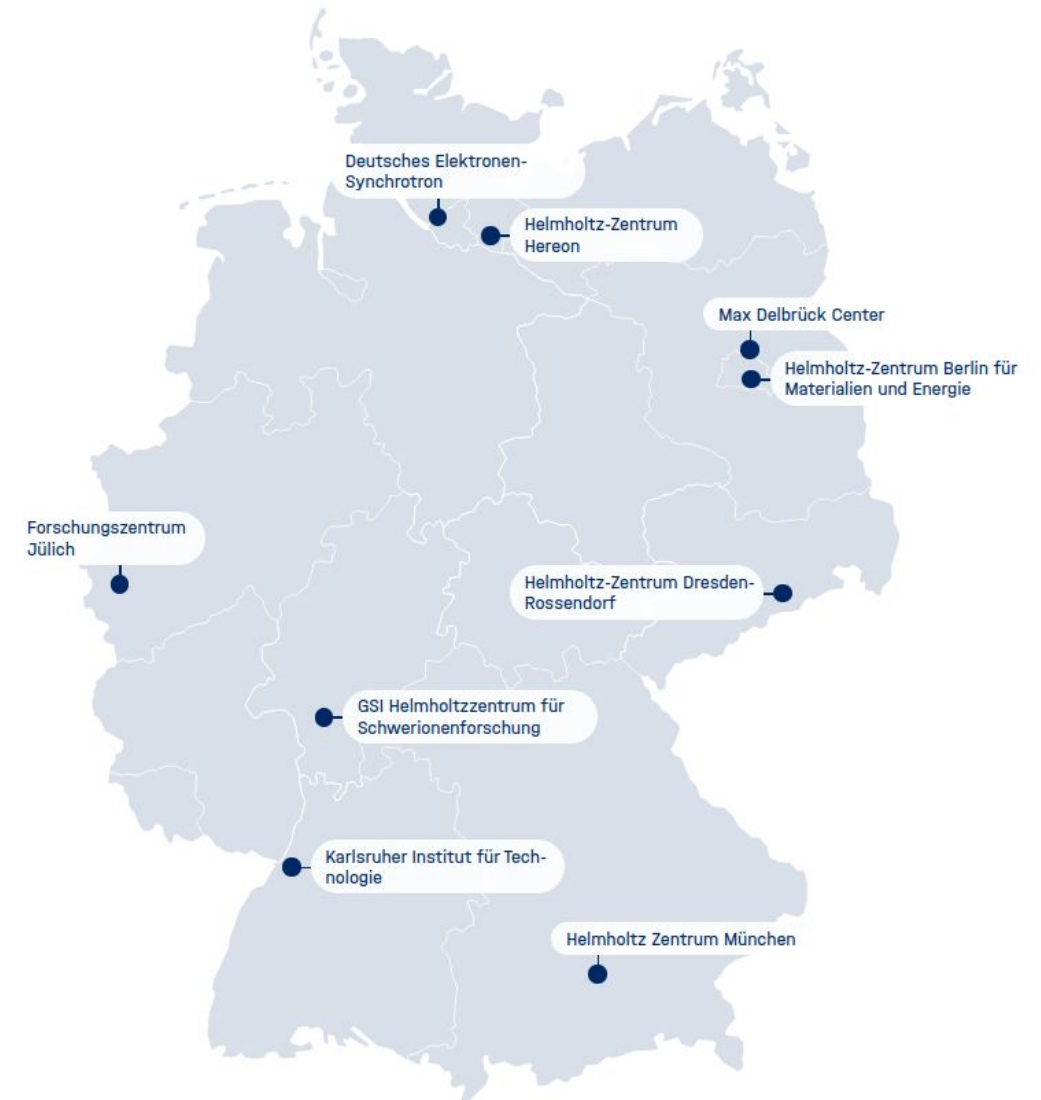
Dr. Kushal Ramakrishna, Scientific Researcher, HZDR, Germany

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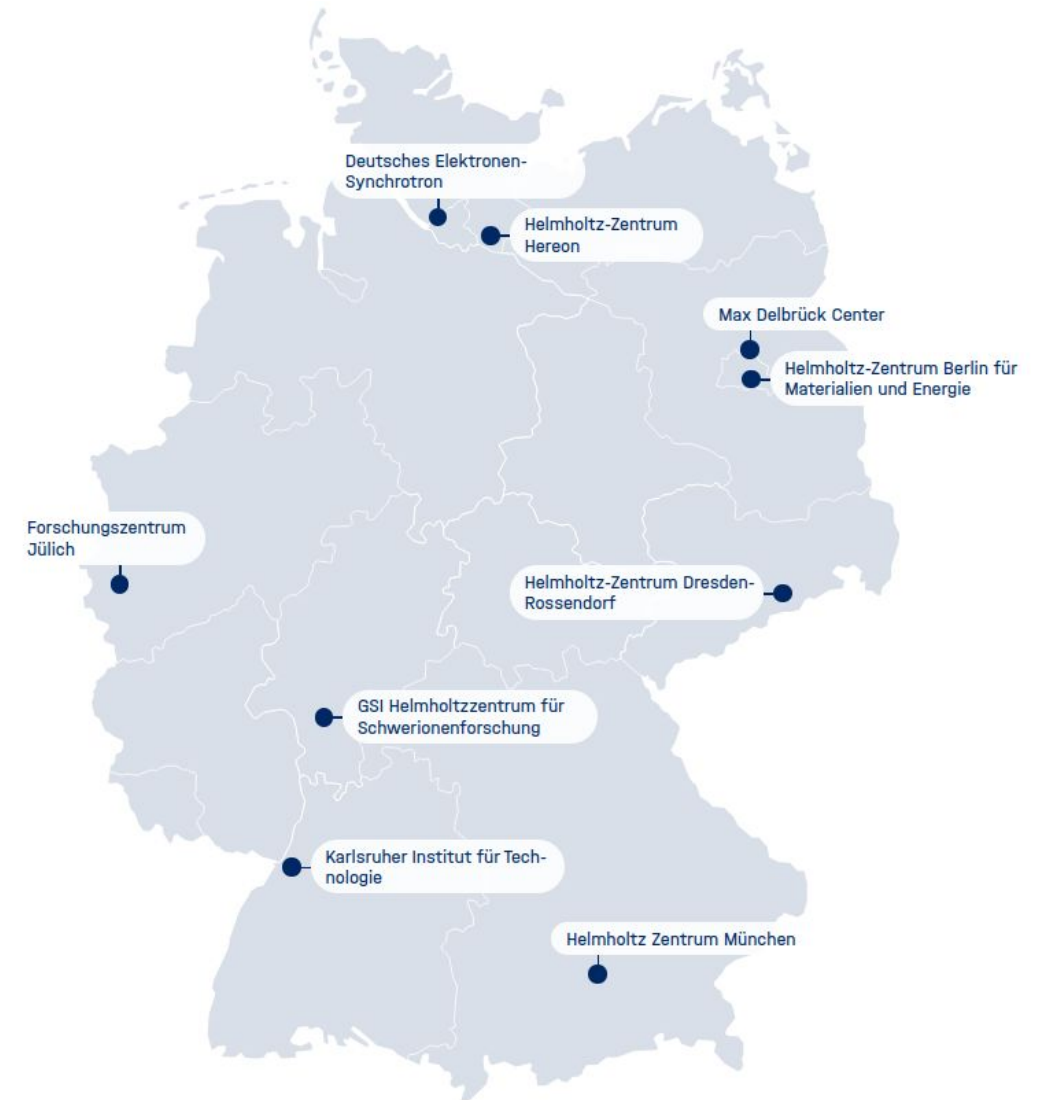
HPC-Gateway

- Enable access → Provide companies with HPC infrastructure, large datasets, and AI expertise.
- Support collaboration → Funded Helmholtz initiative (€18M) across 9 centers, supporting joint projects with industry.
- Build sustainability → Offer expert guidance and training to strengthen AI innovation, knowledge transfer, and Germany's digital competitiveness.



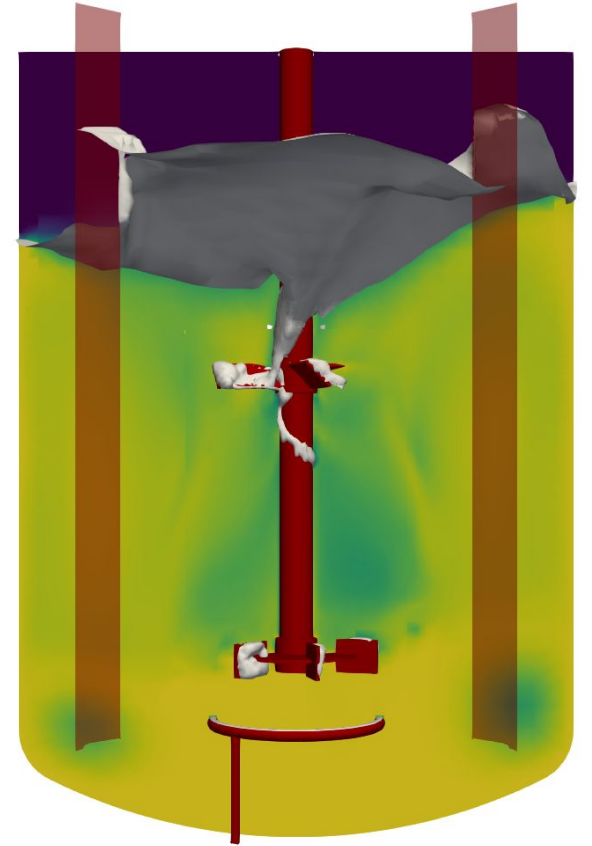
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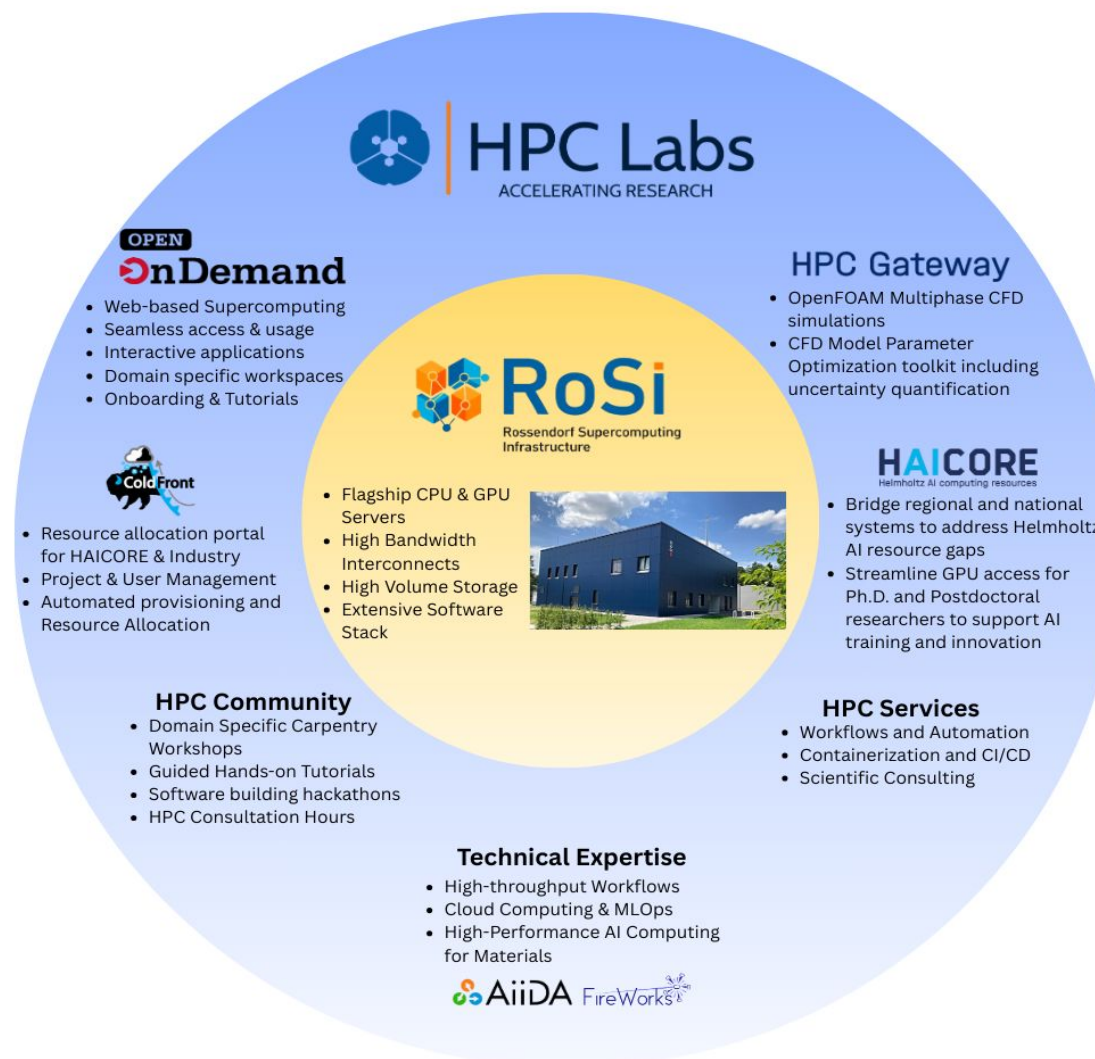
Pilot Project - “Center for industrial multiphase flow competence”

- Aim: To increase the digitalization and efficiency of the development processes in the participating companies through the use of modern simulation methods for multiphase flows.
- Build up a transfer platform for joint activities with industry
- 7 Workpackages with 3 Industry partners:
 - Chemical and process engineering – BASF SE
 - Automotive industry – MAHLE International GmbH
 - Hydrogen production – Sunfire SE



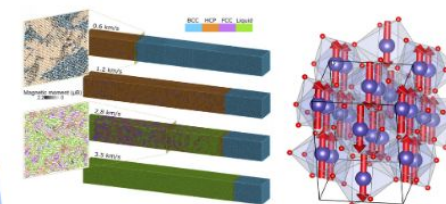
CFD-Simulation of a stirred tank reactor

High Performance Computing Ecosystem at HZDR

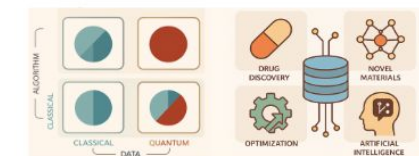


RESEARCH AREAS

Computational Material Science



Quantum Machine Learning



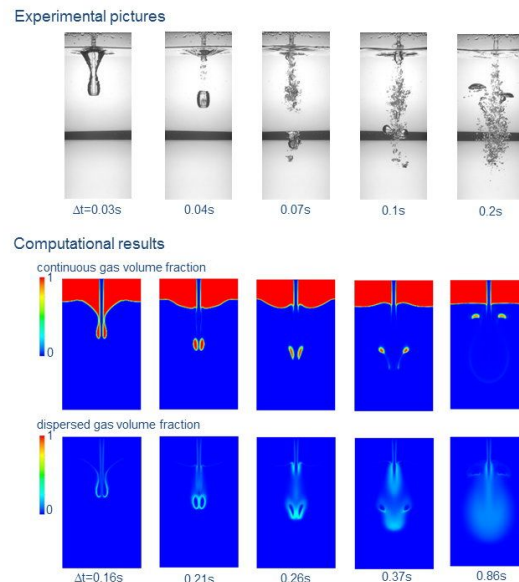
Research Software Development



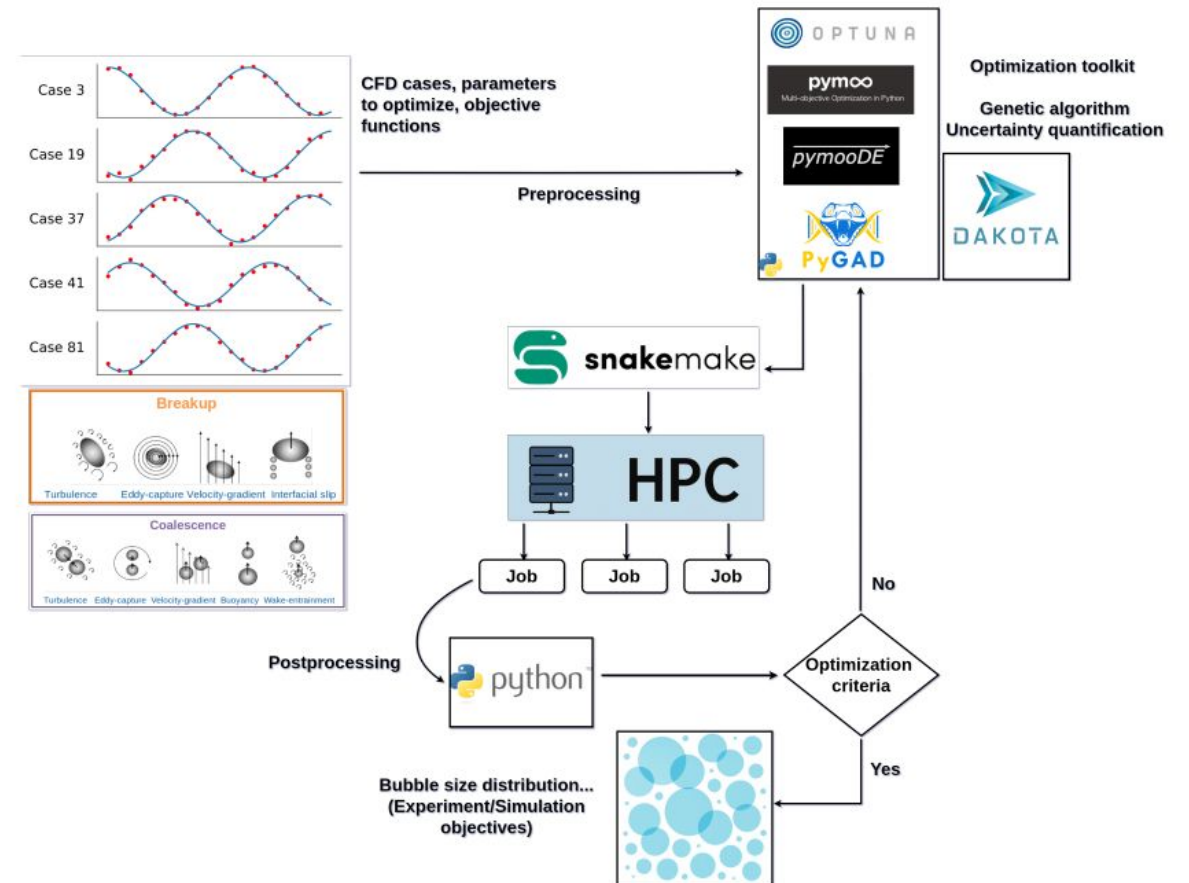
Dr. Henrik Schulz, Head of IT Infrastructure | Mani Lokamani, HPC Group Leader | Varun Sudharshnam, HPC Group | Jens Lasch, HPC Group | Dr. Kushal Ramakrishna, HPC Group

Workpackage – Parameter Optimization Toolkit

- Aim: Optimization of simulation parameters, runtime & resource utilization for CFD Cases.
- Integration of optimization tools in HPC workflows
 - Choose cases, parameter space, objective function criteria, HPC resources.
- Dedicated workspace/application on Open OnDemand portal to customized to user case.



S. Hänsch et al. *International Journal of Multiphase Flow*, 47 (2012)



CFD FOR PROCESS OPTIMIZATION:

CHALLENGES AND OUR RECOMMENDER SYSTEM APPROACH

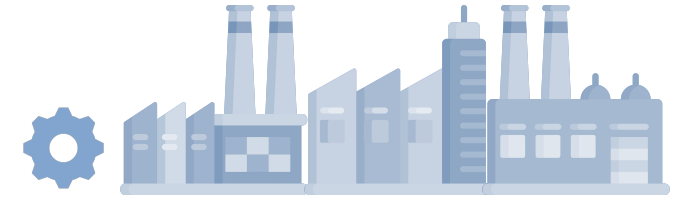
Dr. Susann Hänsch, Post Doctoral Scientific Researcher, HZDR, Germany

Date: September 11, 2025



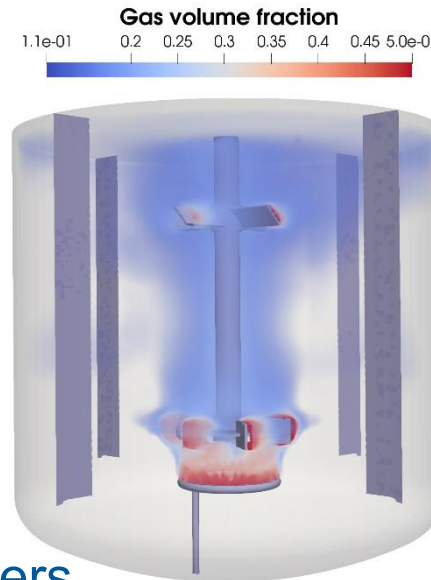
MOTIVATION

CFD FOR THE DIGITALIZATION OF PROCESS OPTIMIZATION

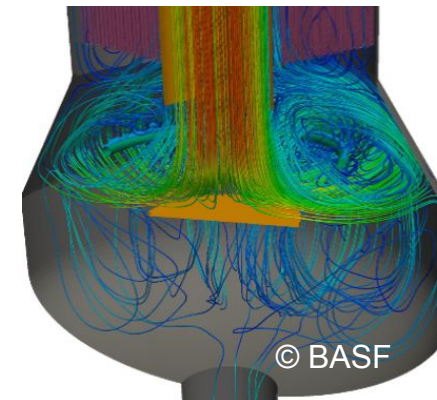


- Computational Fluid Dynamics:
 - Detailed flow analysis
 - Efficient optimization of industry components, and
 - their-upscaling
- Industrial CFD usage limited due to:
 - Model complexity
 - Abundance of options/parameters
 - Constant stream of new models

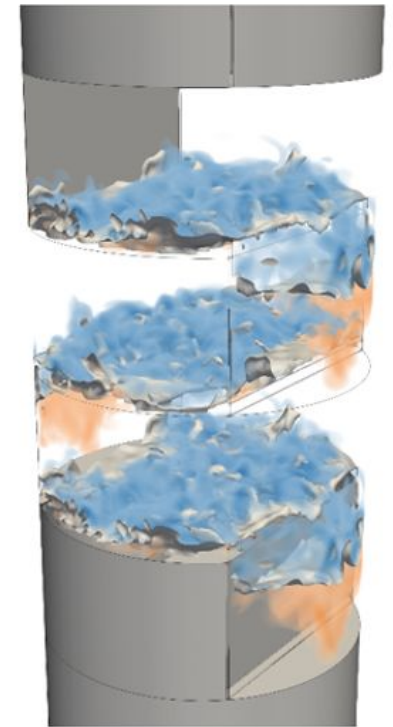
Stirred tanks



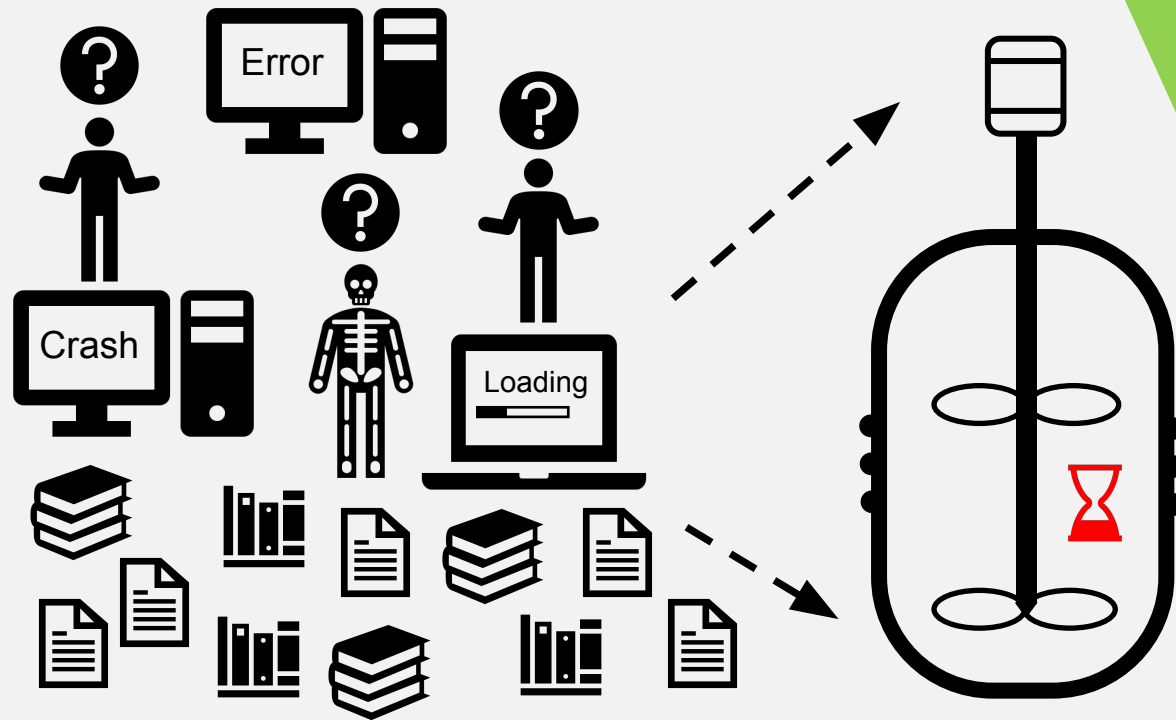
Jet loop reactors



Distillation columns

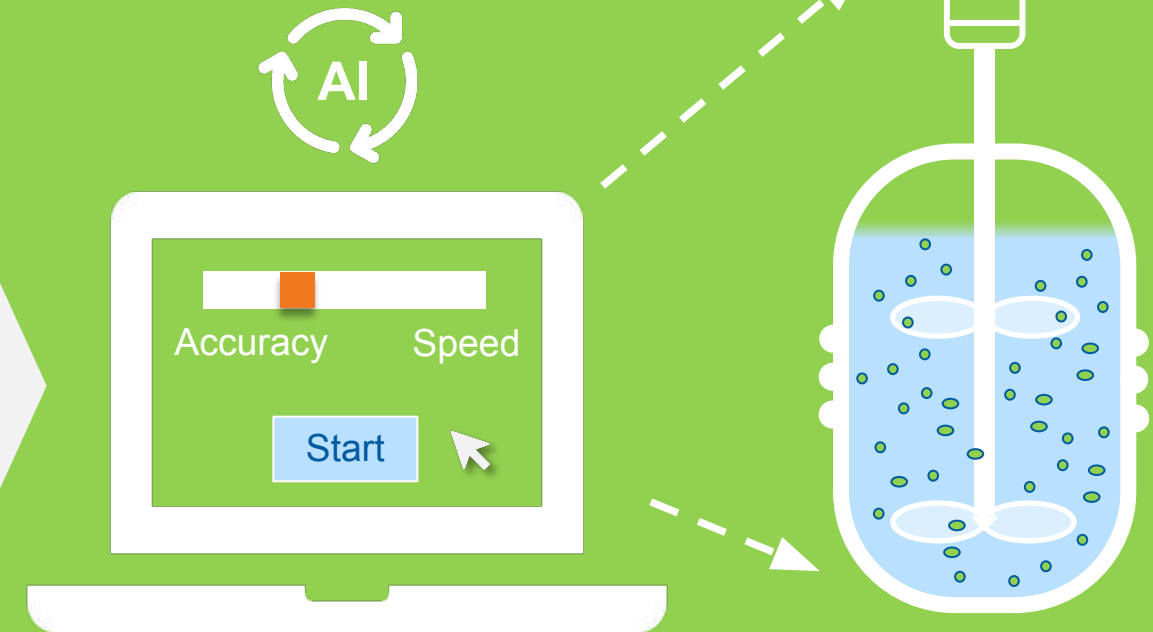


TOWARDS DATA-DRIVEN ENGINEERING



- ✗ Complicated model analyses
- ✗ Time lost on simulation setup
- ✗ Isolated temporary expertise

Recommender System



- ✓ Immediate CFD simulation setup
- ✓ Focus on product development
- ✓ Data preservation + collaboration

PERFORMANCE MATRIX FOR A RECOMMENDER SYSTEM

Closure model testing using 




CFD model portfolio

Select model

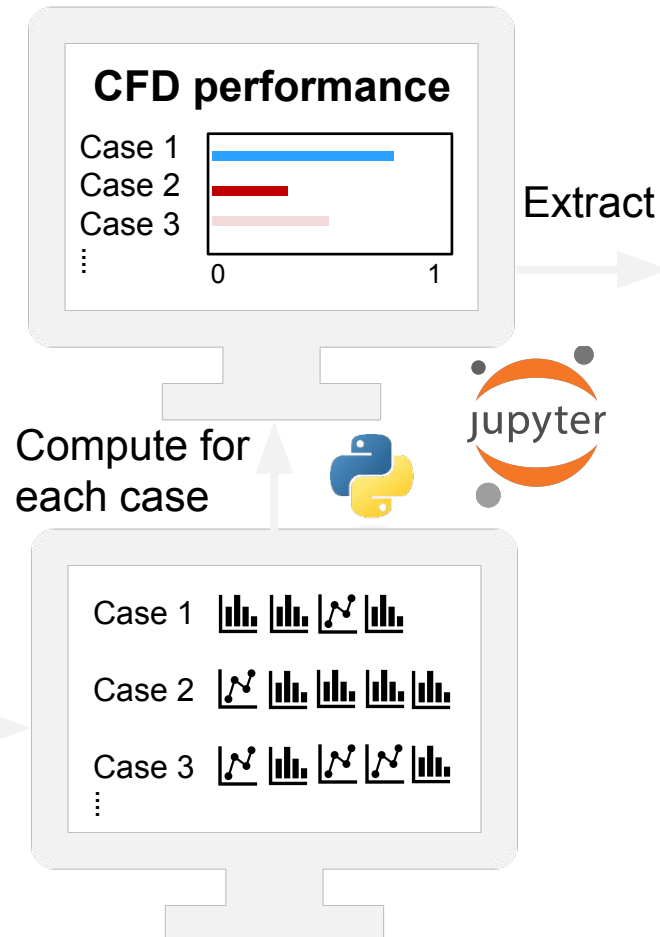


Automation

Batch-processing of CFD cases using the snakemake workflow management 

Cases 
Growing CFD database

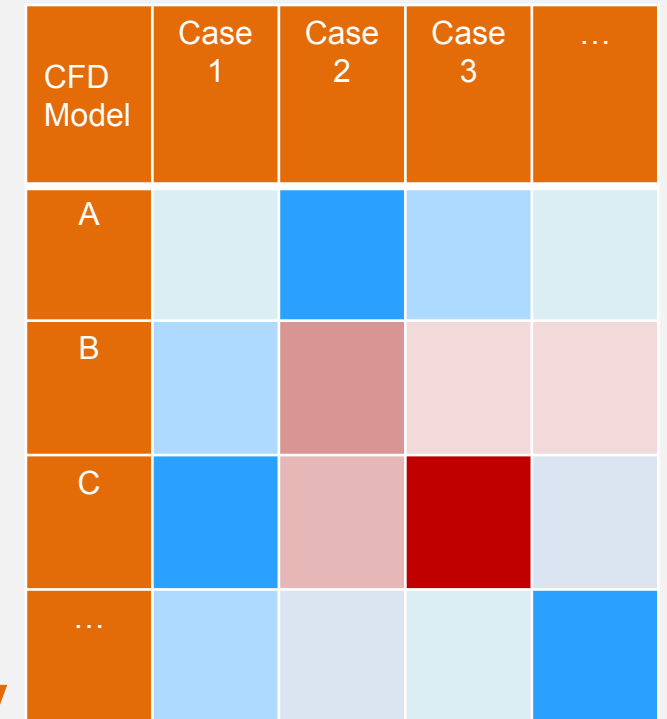
Select cases



Performance matrix

> 700 CFD cases

> 0.5 Mio. model combinations



0 CFD Performance 1

TOMOGRAPHY AND AI-BASED DENOISING:

- ❑ ULTRAFAST CT (ROFEX) AT HZDR
- ❑ USING MACHINE LEARNING TO IMPROVE NOISY DATA
- ❑ IMPACT ON SCIENCE & INDUSTRY APPLICATIONS

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ML-BASED DENOISING OF TOMOGRAPHY DATA

ULTRAFAST ELECTRON BEAM X-RAY COMPUTED TOMOGRAPHY (ROFEX) AT HZDR

Why it matters:

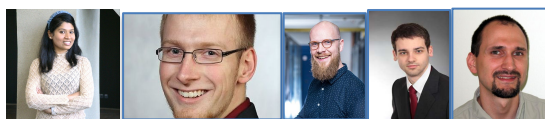
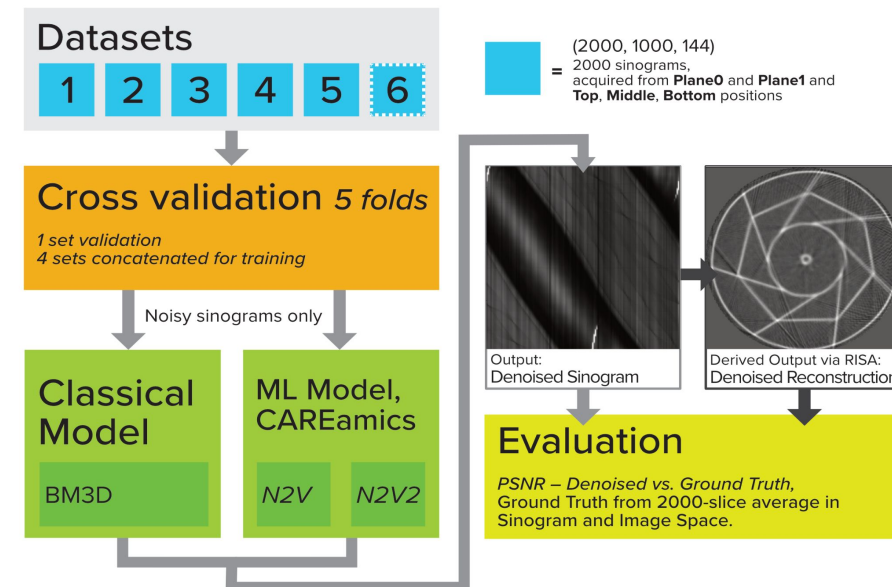
- Tomography data is often noisy.
- Noise limits image quality & reconstruction accuracy.
- Impacts research & industry (healthcare, material science, semiconductors)



What we did:

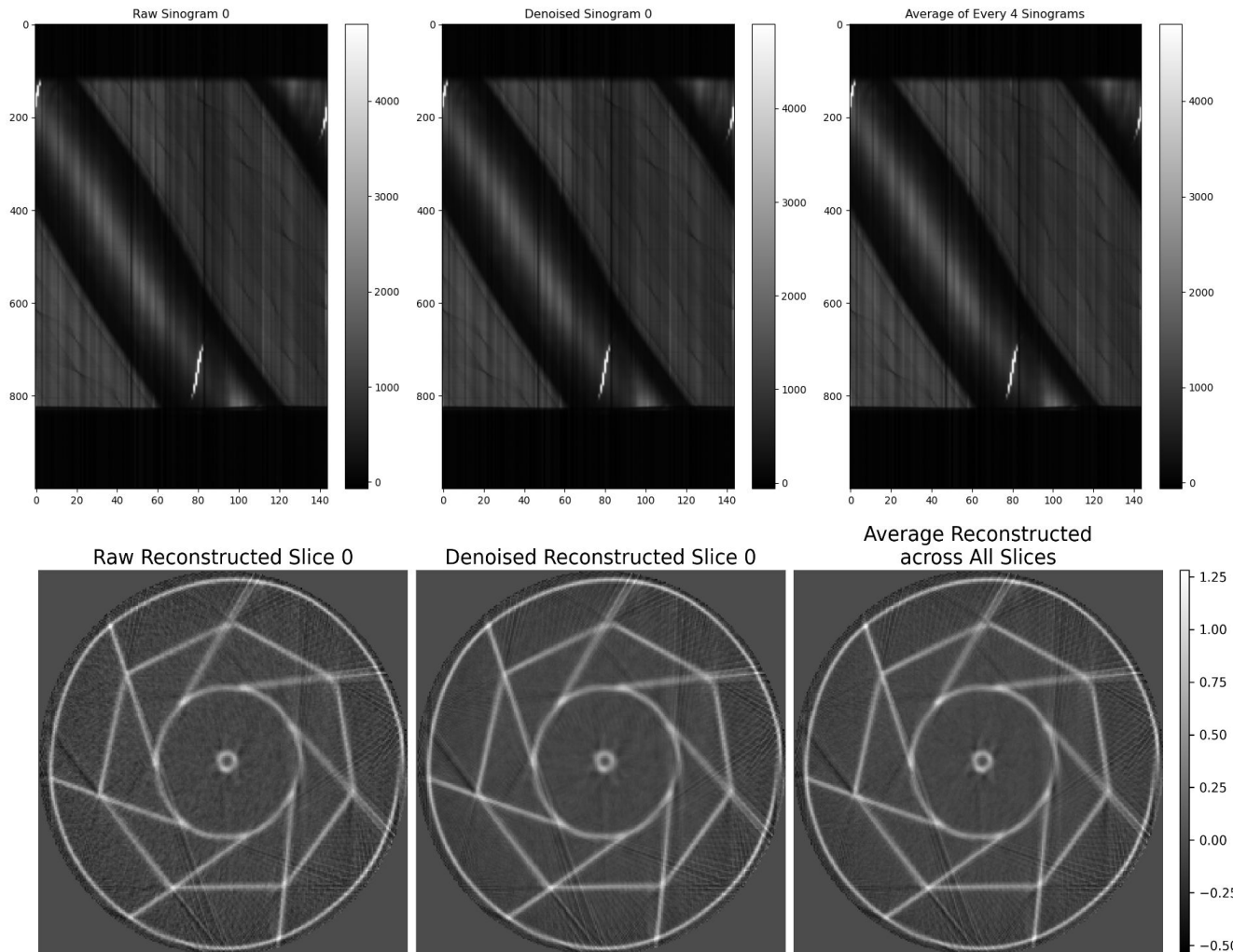
- Applied deep learning on sinogram data
- Compared against non-learning baseline

→ Reconstruction of denoised sinograms led to improved image quality



RESULT AND POTENTIAL IMPACT

IMPROVING IMAGE QUALITY FOR SCIENTIFIC AND INDUSTRIAL APPLICATIONS



- Noise2Void robust^{1,2}
- Deep learning models outperform traditional non-learning methods
- Higher PSNR → Clearer Images
- Working on deep learning architectures for better generalization
- Planned applicability to industrial workflows (microscopy, metrology, semiconductors)

¹Krull, A., Buchholz, T.-O., & Jug, F. (2019). Noise2Void: Learning denoising from single noisy images. In CVPR 2019. [Link](#)

²Batson, J., & Royer, L. (2019). Noise2Self: Blind denoising by self-supervision. In ICML 2019, PMLR 97:524–533. [Link](#)

Thank You!

With special thanks to my collaborators

Building bridges between
HPC/AI research and industry
– from fluid simulations to
real-time tomography.



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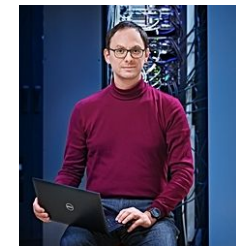
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